IT5020 Social Network Analysis	Assignment - I	Yuvaraj K 2022506079

Social Media Applications

S. No	Name	Type	Features
1	WhatsApp	Traditional	Secure messaging for privacy. One-on-one and group calling. Temporary stories visible for 24 hours. Broadcast updates to followers. Send images, documents, and videos up to 2GB.
2	LinkedIn		Connect with industry professionals. Apply for jobs directly. Courses for skill development. Post articles, updates, and videos. Direct messaging, even to non-connections.
3	Telegram		Access chats from multiple devices. Support for up to 200,000 members. Customizable bots for tasks. Self-destructing encrypted messages. Sends full-quality images & videos.
4	KakaoTalk		Instant chats with free stickers. Integrated mobile payments. Join public chat rooms by interest. Fun voice effects in calls. Set reminders and schedules.
5	BeReal	Media Sharing	Takes a front & back camera shot simultaneously. Prompts users to post within a set time. Focuses on authenticity. Older posts disappear after 24 hours. No public browsing, only for added contacts.
6	Gmail		Categorizes emails (Primary, Promotions, Social). Recall emails within a time limit. Self-destructing and protected emails. Syncs with Drive, Calendar, and Meet. AI-powered text suggestions.
7	Teams		Meetings with up to 1,000 participants. Organized workspaces for discussions. Works with Word, Excel, and PowerPoint. AI-enhanced virtual backgrounds. Connects with third-party services.
8	YouTube		Share content globally. Short-form vertical videos. Earn revenue from content. Follow favourite creators. Auto-generated and manual subtitles.
9	BadmintonCentral	Discussion	Covers professional tournaments. Discussion boards for badminton lovers. Insights on rackets, shoes, and gear. Match breakdowns and predictions. Advice for improving skills.
10	Fandom		Community-built knowledge bases. Discussion spaces for niche topics. Covers movies, games, and TV shows. Speculation and breakdowns. Easily share articles.
11	Discord		Organized communication within servers. Live-stream gaming or presentations. Custom bots for moderation and games. Communities for every interest. Custom emojis, HD streaming, and perks.
12	Reddit		Discussion groups for every topic. Community-driven content ranking. Reward valuable posts and comments. Exclusive Q&A sessions with experts. Filter content based on preferences.
13	Screen Rant	Social Blogging	In-depth analysis and ratings. Entertainment industry updates. Movie breakdowns. Conversations with filmmakers & actors. Editorials on movies, comics, and pop culture.
14	CBR		Covers Marvel, DC, and manga. Compares superheroes & villains. Features with comic creators & artists. Covers superhero films & shows. Deep dives into comic lore.
15	OneFootball		Real-time scores and commentary. Latest player movements. Goals and key moments. News tailored to favourite teams. Watch games directly in-app.
16	This is Anfield		Exclusive club coverage. In-depth analysis of upcoming and past games. Exclusive stories from players. Breakdown of Liverpool's playing style. Articles written by Liverpool supporters.
17	Google Maps	Review	User feedback on locations. Real-world location previews. Live navigation assistance. Downloadable directions without the internet. Images from visitors.
18	Justdial		Find local shops and services. User opinions on businesses. Appointments for plumbers, salons, etc. Contact businesses directly. Deals on services and restaurants.
19	Turf Town		Reserve football and cricket grounds. Feedback on turf quality. Find players for your game. Discounts for frequent bookings. Scorekeeping for local matches.
20	IMDb		User and critic reviews. Full credits for every film & show. Upcoming movie previews. Personalized recommendations. Film earnings and trends.

Social Network Tools

S. No	Name	Type	Features
1	KeyLines	Commercial	Advanced graph visualization for large datasets. Real-time data analysis and filtering. Customizable UI with various layouts.
2	Sentinel Visualizer		Link analysis for investigative intelligence. Multi-layered network visualization. Integrated geospatial mapping features.
3	Unicet		Focuses on criminology and security networks. Social behaviour pattern recognition. Graph-based risk assessment tools.
4	Maltego		Cyber threat intelligence and forensics. Automated data mining and link analysis. Integrates with OSINT and security tools.
5	GraphXR		3D graph visualization for deep network insights. Supports multi-dimensional datasets. Real-time collaborative analysis.
6	NetMiner		Big data analytics with machine learning integration. Dynamic social network modelling. Interactive network visualization.
7	Pajek Pro		Designed for large-scale network analysis. Advanced clustering and partitioning. Efficient for social and biological networks.
8	Linkurious Enterprise		Graph analytics for fraud and cybersecurity detection. Intuitive web-based interface. Seamless integration with Neo4j and other databases.
9	SocioSpyder		Real-time social media monitoring. Sentiment analysis and trend detection. Automated report generation.
10	Tom Sawyer Perspectives		Enterprise-level graph visualization. Supports massive data sets with optimized rendering. Custom workflow automation for analytics.
11	Gephi	Freeware	Interactive graph exploration and visualization. Supports large networks with high performance. Modular system with plugins for added functionality.
12	NodeXL Basic		Excel-based network analysis tool. Built-in metrics for centrality and clustering. Supports import from social media data sources.
13	Graphviz		Automated layout generation for graphs. Supports multiple file formats for input/output. Customizable visualization styles.
14	NetworkX		Python library for complex network analysis. Wide range of built-in algorithms. Handles both small and large graphs efficiently.
15	Cytoscape		Ideal for biological and biomedical network analysis. Plugin-based architecture for extended features. Supports multiple graph file formats.
16	Tulip		Scalable graph visualization framework. Extensible with Python and C++ APIs. Advanced clustering and filtering techniques.
17	Visone		User-friendly interface for network visualization. Statistical analysis of social network structures. Dynamic graph analysis over time.
18	SocNetV		Easy-to-use network visualization and analysis. Calculates various SNA metrics. Supports multiple graph file formats.
19	ORA-Lite		Organizational risk assessment through network analysis. Identifies key players in networks. Predictive modelling for decision-making.
20	JUNG		Java-based flexible graph processing. Customizable visualizations with various layouts. Extensive library for algorithms and data structures.
21	Netlytic	Shareware	Cloud-based social media and text network analysis. Supports Twitter, YouTube, and other platforms. Automated text clustering and sentiment analysis.
22	UCINET		Comprehensive tool for SNA calculations. Extensive network visualization options. Supports matrix-based data input.
23	Polinode		Organizational network analysis with interactive dashboards. Identifies key influencers in teams. Supports CSV and survey-based data import.
24	Kumu		Visual storytelling with interactive network maps. Relationship mapping for systems thinking. Supports live collaboration.
25	SNAP		Optimized for large-scale network datasets. Provides fast and efficient graph algorithms. Supports dynamic network analysis.
26	EgoNet		Focuses on personal network data collection. Analyses individual network structures. Supports interview-based data entry.
27	Graphistry		GPU-accelerated graph visualization. Real-time interaction with large datasets. Integrates with Python and security tools.
28	Relation Browser		Dynamic visualization of social relationships. Interactive filtering and exploration. Timeline-based network evolution analysis.
29	Power BI		Integrates Power BI with graph databases. Interactive network visualizations. Supports advanced filtering and data queries.
30	SNA Toolkit		Comprehensive network metric calculations. Supports dynamic network evolution tracking. Compatible with various data sources.

Measures Pertaining to Node-Wise Analysis

S. No	Measure Name	Formula	Description
1	Degree	$d(v) = \sum A_{vu}$	Number of direct connections a node has.
2	Betweenness	$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$	Measures how often a node appears on shortest paths between other nodes.
3	Closeness	$C_C(v) = \frac{1}{\sum d(v,u)}$	Inverse of the average shortest path distance from the node to all others.
4	Eigenvector Centrality	$x_i = \frac{1}{\lambda} \sum_j A_{ij}\, x_j$	Importance of a node based on its connections to other important nodes.
5	Katz Centrality	$x_i = \alpha \sum_j A_{ij}\, x_j + \beta$	Generalized eigenvector centrality that accounts for indirect connections.
6	PageRank	$PR(v) = \frac{1-d}{N} + d \sum_{u \in N_v} \frac{PR(u)}{d(u)}$	Google's algorithm to rank nodes based on incoming links.
7	Hub Score	$h_i = \sum_j A_{ij}\, a_j$	Measures node influence in a network by treating it as a "hub" linking to authorities.
8	Authority Score	$a_i = \sum_j A_{ji}\, h_j$	Measures the credibility of a node based on how many high-quality "hubs" point to it.
9	Local Clustering Coefficient	$C(v) = \frac{2T(v)}{d(v)(d(v)-1)}$	Measures the tendency of a node's neighbours to form a clique.
10	Global Clustering Coefficient	$C = \frac{\text{number of closed triplets}}{\text{number of all triplets}}$	Probability that a randomly selected triplet of nodes is closed.
11	Bridging Centrality	$BC(v) = \frac{d(v)}{\sum d(u)} \times C_B(v)$	Identifies nodes that act as bridges between clusters.
12	Eccentricity	$E(v) = \max d(v,u)$	Maximum shortest path from the node to any other node.
13	Radiality	$R(v) = \sum \left(D_{\max} - d(v,u)\right)$	Measures how close a node is to all other nodes in the network.
14	Percolation Centrality	$PC(v) = C_B(v) \times P(v)$	Measures a node's importance while considering its failure probability.
15	Decay Centrality	$DC(v) = \sum_u \delta^{d(v,u)}$	A discount factor-based measure that considers influence over distances.
16	Harmonic Centrality	$H(v) = \sum \frac{1}{d(v,u)}$	Variation of closeness centrality that avoids issues with disconnected nodes.
17	Participation Coefficient	$P(v) = 1 - \sum \left(d_c(v)/d(v)\right)^2$	Measures how evenly a node's links are distributed among different communities.
18	Coreness	k-core where $d(v) \geq k$	Identifies the largest subgraph in which all nodes have at least k connections.
19	Accessibility	$A(v) = \sum_{k=1}^{\infty} P_k\,(v) e^{S_k(v)}$	Measures the effective reach of a node in a network.
20	Flow Betweenness	$FB(v) = \sum_{s,t} f_{st}\,(v)$	Similar to betweenness but considers actual flow (e.g., information or traffic).
21	Laplacian Centrality	$LC(v) = x_v^T L x_v$	Quantifies a node's impact on the network's structure.
22	Geodesic Vulnerability	$GV(v) = \sum_{s,t} \frac{\sigma_{st} - \sigma_{st}(v)}{\sigma_{st}}$	Measures how much the shortest paths are affected when a node is removed.
23	Centroid Value	$CV(v) = \sum_u \mid d(u) - d(v) \mid$	Measures how different a node's degree is compared to its neighbours, indicating its relative importance.
24	Structural Diversity	$SD(v) = \mid C(v) \mid$	Counts the number of distinct communities a node belongs to, showing its cross-cluster connections.
25	Weighted Degree	$WD(v) = \sum w_{vu}$	Similar to degree but accounts for weighted connections.

Commercial Social Network Profiles

S. No	Music	Film	Television	Public Figures	Events
1	@a24music	@a24	@adultswim	@wernerherzog_official	@fantasiafest
2	@flyinglotus	@neonrated	@criterionchannel	@bjork	@sxsw
3	@ninja_tune	@criterioncollection	@davelyonsonfilm	@neiltyson	@sundanceorg
4	@deadbeats	@rygus	@britboxtv	@mirandakerr	@venezia78
5	@ostinato_records	@pattinson360	@serialpodcast	@patti_smith_hq	@torontofilmfest
6	@stonesthrow	@thewitchmovie	@twinpeaksarchve	@rickrubin	@cannes_film_festival
7	@worldcircuit	@mandy_network	@blackmirrorfans	@guillermodeltorofilm	@paxgaming
8	@subpop	@cinemawithoutborders	@horrorhosttv	@vincentcasselofficial	@meltfestival
9	@kexp	@kinolorber	@darknetflixde	@janellemonae	@moogfest
10	@anoushkashankar	@noahbaumbachfan	@misterrogerssay	@sarahpolleyfilm	@theweekndexp
11	@sunn_o)))	@whokilledcaptainalex	@midnightsocietytv	@tylertheshadowgov	@shambhala_mf
12	@ghostmane	@jodorowskysdune	@altcomedyhub	@leehyunwoo	@beyondwonderland
13	@deathgripz	@drafthousefilms	@venturebrosarchive	@mfdoomforever	@sunburnfestival
14	@boilerroomtv	@weirdhorrormovies	@quebecois_tv	@agnesvarda_forever	@tokyogame_show
15	@bandcamp	@darkskyfilms	@scifichannellive	@koreeda_hirokazu	@comic_con
16	@rnsound	@bonghive	@lost_fans_unite	@icecubeclassic	@berlinale
17	@hyperdubrecords	@agnèsvardaofficial	@netflixnordic	@keanuuniverse	@rollingcloud
18	@erasedtapes	@gasparnoefilms	@thelastdrivein	@officialjohncandy	@wffwhistler
19	@noise_music_hub	@slowcinema	@hbohighbrow	@yokoonoofficial	@midemofficial
20	@glastonburyfest	@letterboxd	@crunchyroll	@hanszimmerlive	@horrorthon

Top 10 YouTube Channels – Region Wise

India			Australia			France		
Name	S	YOE	Name	S	YOE	Name	S	YOE
T-Series	287M	14	BouncePatrol	32.6M	11	Tibo InShape	26.1M	11
SET India	182M	15	Chloe Ting	25.6M	8	SQUEEZIE	19.4M	13
Zee Music Company	114M	10	How Ridiculous	23.5M	15	Lofi Girl	14.8M	5
Goldmines Telefilms	103M	6	LazarBeam	22.8M	10	GIMS	12.7M	11
Sony SAB	99.4M	14	CKN	19.1M	9	Mayamystic	11.5M	8
ChuChu TV	95.1M	12	HowToBasic	17.7M	13	Soolking Official	10.8M	6
Zee TV	88M	15	Lachlan	15.3M	11	Michou	10.3M	9
Colors TV	78.9M	14	GLITCH	13.1M	4	Masha et Michka	10.1M	8
T-Series Bhakti Sagar	74.1M	13	Jiemba Sands	12.7M	9	Lama Fache	9.77M	8
Tips Official	73M	17	SR Toys Collection	12.1M	10	Tuvok12	9.65M	2

United States of America			Spain			United Kingdom		
Name	S	YOE	Name	S	YOE	Name	S	YOE
MrBeast	366M	13	Mikecrack	54.7M	9	Peppa Pig	38.7M	11
Cocomelon	190M	18	invictor	29.3M	11	NoCopyrightSounds	33.9M	11
Vlad and Niki	134M	6	AuronPlay	29.1M	15	DanTDM	29.2M	12
Kids Diana Show	131M	9	Chuster	26.3M	7	Coldplay	27.5M	16
Like Nastya	126M	8	Las Ratitas	25.6M	9	Dan Rhodes	26.9M	9
Stokes Twins	114M	7	ElTrollino	22.2	11	Miss Katy	25.3M	10
5-Minute Crafts	81.1M	8	TheGrefg	19M	13	KSI	25M	15
Alan’s Universe	80.8M	3	Luli Pampin	18.7M	6	Mister Max	24.9M	10
Baby Shark	80.5M	10	Alextis	18.5M	1	Dua Lipa	24M	9
A4	75.3M	10	Willyrex	18.3M	14	Jelly	23.5M	10

S – Subscriber Count
YOE – Years of Existence